Linear models cannot explain several important features common to financial data:

- leptokurtosis or "fat tails": high negative returns that have higher-than-expected probability in historic time-series;
- volatility clustering or volatility "pooling": big shocks tend to follow big shocks (either direction) and small shocks tend to follow small shocks;
- leverage effects or volatility "asymmetry": for example, volatility of a stock price tends to increase when its price drops (Black, 1976).

In the (invertible) ARMA model:

$$y_t = a_0 + \sum_{j=1}^{p} a_1 y_{t-j} + \sum_{j=1}^{q} b_q \varepsilon_{t-j} + \varepsilon_t,$$

$$y_t = \underbrace{g_t(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)}_{\text{conditional mean}} + \nu_t \underbrace{h_t^{\frac{1}{2}}(\varepsilon_{t-1}, \varepsilon_{t-2}, \dots)}_{\text{conditional variance}},$$

Auto-Regressive Conditional Heteroscedasticity

Simplest example was proposed by Engle (1982):

$$h_t \equiv \mathsf{E}(\varepsilon_t^2 \,|\, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots) = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2.$$

The ARCH(1) can be easily extended to an ARCH(q) process:

$$\begin{aligned} h_t \equiv \mathsf{E}(\varepsilon_t^2 \,|\, \varepsilon_{t-1}, \varepsilon_{t-2}, \dots) &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \alpha_2 \varepsilon_{t-2}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2, \\ &= \alpha_0 + \alpha(L) \varepsilon_{t-1}^2, \end{aligned}$$

The Generalized ARCH model (proposed by Bollerslev, 1986) is:

$$\begin{aligned}
\varepsilon_t^2 &= \nu_t^2 h_t, \\
h_t &= \alpha_0 + \sum_{j=1}^{q} \alpha_j \varepsilon_{t-j}^2 + \sum_{j=1}^{p} \beta_j h_{t-j}, \\
&= \alpha_0 + \alpha(L)\varepsilon_{t-1}^2 + \beta(L) h_{t-1},
\end{aligned}$$

known as GARCH(p, q).

The presence of ARCH errors does not invalidate OLS, but there exist more efficient non-linear estimators to estimate conditional-mean coefficients.

Breusch-Pagan test for heteroscedasticity: $H_0 : \alpha_1 = \cdots = \alpha_q = 0$, $H_1 :$ ARCH(q).

1. Estimate the conditional mean using OLS and save residuals in $\{\widehat{\varepsilon}_t\}$.
2. Regress $\widehat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \widehat{\varepsilon}_{t-1}^2 + \cdots + \alpha_q \widehat{\varepsilon}_{t-q}^2$ and compute R_ε^2.
3. Test using $\text{LM}_{\text{ARCH}} \equiv T R_\varepsilon^2 \sim \chi_q^2$.

Testing can provide a diagnostic tool, which is useful in some contexts.

If the objective is to forecast volatility, then it is always relevant to extend a given class of models (e.g. ARMA or ARDL) to specifications with heteroscedastic errors.